

CBSE Class 10 Mathematics

Time Allowed: 3 Hours

Maximum Marks: 30

General Instructions:

1. This question paper contains 12 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A bag contains 15 cards numbered from 1 to 15. One card is drawn at random from the bag. The probability that the number on the drawn card is a prime number is
(A) (A) $\frac{1}{3}$
(B) (B) $\frac{2}{5}$
(C) (C) $\frac{7}{15}$
(D) (D) $\frac{1}{5}$
2. For a given grouped frequency distribution, the assumed mean method formula is $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$. If the assumed mean $a = 45$, $\sum f_i d_i = 120$, and $\sum f_i = 30$, then the mean $\bar{x}$ is:
(A) (A) 44
(B) (B) 48
(C) (C) 49
(D) (D) 50
3. Assertion (A): The probability of an impossible event is 0. Reason (R): For any event E , the probability $P(E)$ satisfies $0 \leq P(E) \leq 1$.
(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
4. Assertion (A): For a given grouped frequency distribution, the class mark x_i of a class interval is calculated as the average of its lower limit and upper limit. Reason (R): The mean of grouped data is calculated using the formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i represents the class mark of the i^{th} interval.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

1. A box contains 40 cards numbered from 1 to 40. If a card is drawn at random from the box, find the probability that the number on the card is a multiple of 5.
2. For a given frequency distribution, the sum of frequencies $\sum f_i = 50$ and the sum of the product of frequencies and class marks $\sum f_i x_i = 1750$. Find the mean of the distribution.

Section C: Short Answer (3 marks each)

1. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number, (ii) a perfect square number.
2. The following table shows the daily pocket allowance of 50 children in a locality. Find the mean daily pocket allowance using the direct method. — Daily allowance (in Rs.) — 100 — 120 — 120 — 140 — 140 — 160 — 160 — 180 — 180 — 200 — — : — — : — : — : — : — : — : — — Number of children — 12 — 14 — 8 — 6 — 10 —

Section D: Long Answer (4-5 marks each)

1. A box contains a total of 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number, (ii) a perfect square number, and (iii) a number divisible by 5.
2. The following table shows the daily pocket money received by 50 students in a locality. If the mean pocket money is Rs.18, find the missing frequencies f_1 and f_2 given that $f_1 + f_2 = 12$. The distribution of pocket money (in Rs.) is: Class Interval: 11 – 13, 13 – 15, 15 – 17, 17 – 19, 19 – 21, 21 – 23, 23 – 25 Frequency: 3, 6, 9, f_1 , f_2 , 5, 4

Section E: Case Study (4 marks each)

1. A box contains a total of 60 cards numbered from 1 to 60. Some cards are marked with red ink, some with blue ink, and some with green ink. The number of red cards is 12 more than the number of green cards, and the number of blue cards is 8. A card is drawn at random from the box.
 - (i) If the number of green cards is x , find the value of x .

- (A) (A) 15
 (B) (B) 20
 (C) (C) 25
 (D) (D) 10
- (ii) Find the probability that the card drawn is a red card.
- (iii) What is the probability that the card drawn is NOT a blue card?
 (A) (A) $1/15$
 (B) (B) $2/15$
 (C) (C) $13/15$
 (D) (D) $14/15$
- (iv) Find the probability that the card drawn is either green or blue.
2. A survey was conducted by a group of students as part of their environmental awareness project, in which they collected data regarding the number of plants in 20 houses in a locality. The data is summarized in the following frequency distribution table:
- | | | | | | | | | |
|---------------------------------|-------|-------|-------|-------|--------|---------|---------|-----------------------------------|
| — Number of plants — | 0 — 2 | 2 — 4 | 4 — 6 | 6 — 8 | 8 — 10 | 10 — 12 | 12 — 14 | |
| — — : | — — : | — — : | — — : | — — : | — — : | — — : | — — : | Number of houses (f_i) |
| — 1 — 2 — 1 — 5 — k — 2 — 3 — | | | | | | | | The total number of houses is 20. |
- (i) What is the value of the missing frequency k ?
 (A) (A) 4
 (B) (B) 5
 (C) (C) 6
 (D) (D) 7
- (ii) Find the class mark (x_i) for the class interval 8 — 10.
- (iii) Which method is most suitable to calculate the mean of this grouped data?
 (A) (A) Direct Method
 (B) (B) Assumed Mean Method
 (C) (C) Step Deviation Method
 (D) (D) All are equally efficient
- (iv) Calculate the mean number of plants per house.